Impact of Frame Duration on Cross-Channel Interference and Coexistence in IEEE 802.11bn NPCA

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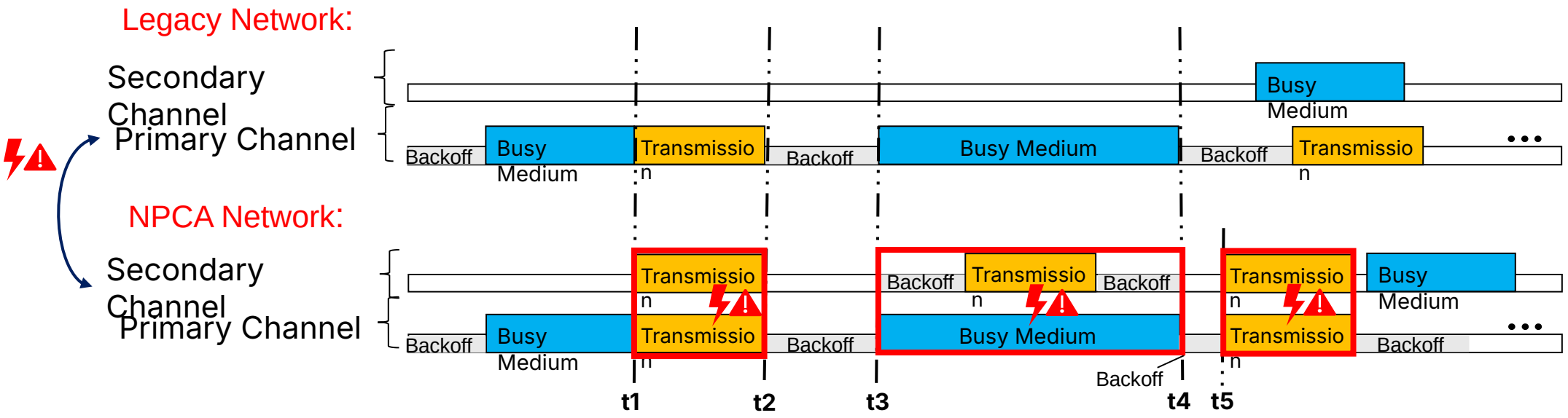
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Outline

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- System Description
- Simulation Setting
- Channel Utilization Analysis
- Throughput Performance
- Conclusion and Future Plans

Motivation1

- Wi-Fi 8 expands channel bandwidth up to 320 MHz to achieve higher data rates.
- However, these wider channels increase the likelihood of Overlapping BSS(OBSS), which intensifies interference in OBSS environments and makes primary channel operation inefficient.
- Non-Primary Channel Access (NPCA) allows STAs to use secondary channels to improve efficiency, but in OBSS environments this may overlap with a neighboring BSS's primary channel and cause harmful interference.



Motivation2

- NPCA is actively discussed in standardization groups and frame duration configuration remains undefined.
- Frame duration remains an open issue in NPCA standardization, and its performance impact is still insufficiently investigated.
- We focus on analyzing how different frame durations influence NPCA operation and coexistence performance.

Contribution

- We analyzed the effect of NPCA frame duration on cross-channel utilization and throughput between BSSs.
- We developed a slot-based simulation framework that evaluates coexistence under different frame lengths.
- We implemented a simulation framework that analyzes the impact of frame duration on channel utilization and throughput.
- Our findings provide insights for NPCA configuration and contribute to Wi-Fi 8 standardization.

System Description

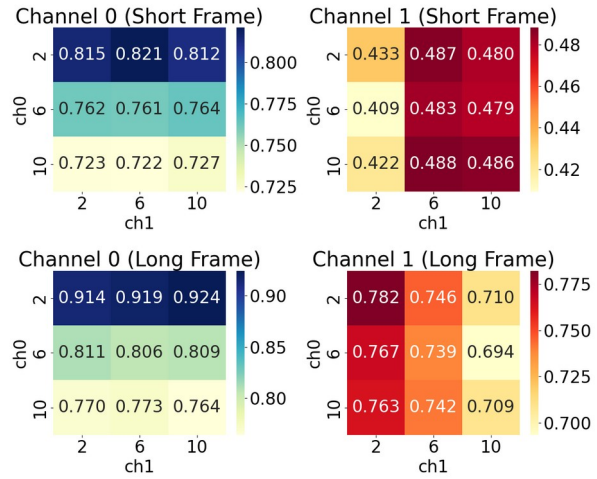
- Two BSS types are considered: Legacy-only and NPCA-enabled
- NPCA activation condition: Primary OBSS is active while the secondary channel is idle
- Two frame duration settings are evaluated: Short and Long
- Performance is measured in terms of channel utilization and throughput.

Simulation Setting

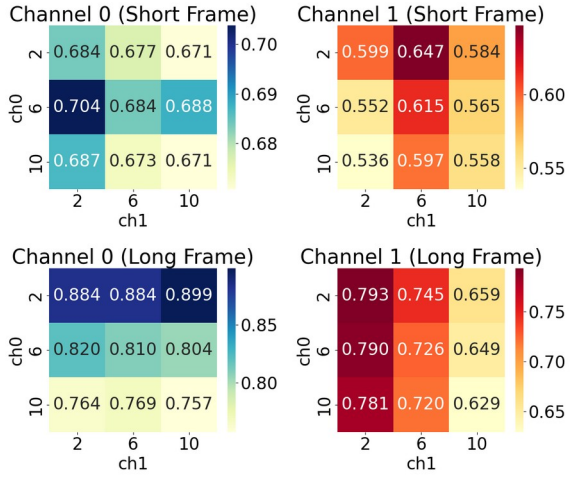
Parameter	Value
Simulation time	500,000μs
Channels	2 × 20 MHZ (Ch0,Ch1)
STAs per channel	2,6,10
NPCA enabled channel	Channel 1 only
OBSS enabled channel	Channel 1 only
Frame duration(Short)	33 slots (300μs)
Frame duration(Long)	165 slots (1,500μs)
Transmission protocol	CSMA/CA

Channel Utilization Analysis

- Channel utilization refers to the valid channel occupation ratio.
  - In Legacy-only operation, all STAs transmit without NPCA and show stable utilization in both channels.
  - With NPCA enabled, STAs in Channel 1 may access Channel 0 during OBSS periods, which introduces interference and degrades its utilization.
- Under (b), short frames intensify interference and reduce utilization on Channel 0, while long frames ease contention but yield only limited NPCA gains.
- NPCA can improve efficiency, but it requires careful configuration to avoid degrading neighboring BSS performance.



(a) Legacy-only

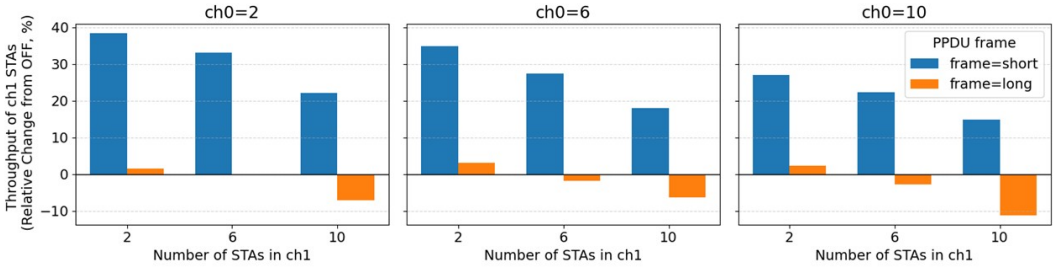


(b) NPCA-enabled

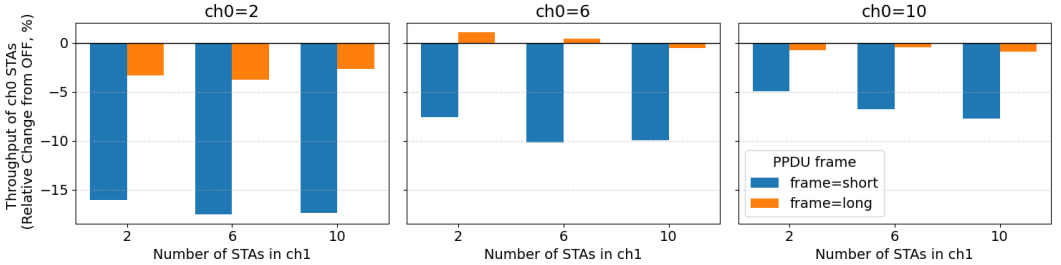
Throughput Performance

- With NPCA, Channel 1 throughput increases under NPCA with short frames.
- However, Channel 0 throughput decreases due to interference, and the loss becomes larger with more STAs.
- With long frames, Channel 1 gain diminishes and the NPCA effect becomes negligible.
- Both Channel 0 and Channel 1 show lower throughput with long frames compared to short frames.

Throughput of Channel 1



Throughput of Channel 0



Conclusion and Future Plans

- This paper analyzed the impact of NPCA on channel utilization and throughput under various STA distributions and frame duration settings.
- NPCA improves channel efficiency during OBSS, but its effectiveness depends on STA numbers and frame duration, reflecting a trade-off between efficiency and fairness.
- Proper configuration and standardization are required to prevent performance degradation.
- Future work will investigate adaptive NPCA strategies and Machine Learning–based Real-Time optimization.